

Unit 2: Data Collection, Cleaning & Preprocessing

Complete README Notes (Hinglish)

Introduction

Real-world data kabhi perfect nahi hota.

Problems:

- Missing values
- Wrong values
- Duplicate values
- Different formats
- Noise
- Outliers

Isliye data ko clean aur preprocess karna padta hai before analysis and machine learning.

Topics Covered

1. Sources of Data (Primary and Secondary)
 2. Handling Missing Data
 3. Outliers
 4. Noise
 5. Data Transformation
 6. Normalization
 7. Data Wrangling using Pandas & NumPy
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1. Sources of Data (Primary and Secondary)

What are Data Sources?

Data kaha se aata hai usko data source bolte hain.

Mainly 2 types:

1. Primary Data
 2. Secondary Data
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Primary Data

Meaning

Jab data directly original source se collect kiya jata hai kisi specific purpose ke liye.

Yaani:

- Fresh data
 - Original data
 - Khud collect kiya gaya data
-

Examples of Primary Data

Situation	Primary Data
YouTube Survey	Audience feedback
Hospital	Patient reports
School	Student attendance
Traffic System	Live camera data
E-commerce	Customer reviews

Methods of Collecting Primary Data

1. Surveys

Questions pooch kar data collect karna.

Example:

- Google Forms
 - Online questionnaires
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2. Interviews

Direct baat karke information lena.

Example:

- HR interview
- Phone interview

3. Observation

Behavior observe karke data lena.

Example:

- Mall customer tracking

4. Experiments

Testing karke results record karna.

Example:

- Medicine testing

5. Sensors and IoT Devices

Machine automatically data collect karti hai.

Example:

- Smartwatch
- Temperature sensor

Advantages of Primary Data

- High accuracy
- Fresh data
- Specific purpose ke liye useful
- More reliable

Disadvantages of Primary Data

- Expensive
 - Time consuming
 - More effort required
-

Secondary Data

Meaning

Jo data pehle se kisi aur ne collect kiya ho aur hum usko reuse karein.

Examples of Secondary Data

Source	Example
Government	Census data
Websites	Kaggle datasets
Company records	Sales reports
Social media	Twitter trends
Research papers	Published studies

Sources of Secondary Data

1. Government Reports

Example:

- Census
 - Economic surveys
-

2. Research Papers

Academic studies and journals.

3. Websites and Portals

Example:

- Kaggle
 - Government websites
-

4. Company Databases

Example:

- Employee records
 - Sales history
-

Advantages of Secondary Data

- Cheap
 - Easily available
 - Saves time
 - Large amount of data available
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Disadvantages of Secondary Data

- May be outdated
 - Less accurate
 - May not match exact requirement
-

Primary vs Secondary Data

Feature	Primary Data	Secondary Data
Source	Original	Already collected
Cost	High	Low
Accuracy	High	Moderate
Time	More	Less
Example	Survey	Census report

Real-Life Example

Suppose Swiggy wants to know: "Customers delivery se satisfied hain ya nahi?"

Primary Data

- Customer survey
- Feedback form
- Delivery tracking

Secondary Data

- Old order reports
 - Existing market research reports
-

Mind Trick

Primary Data

“Pehli baar khud collect”

Secondary Data

“Second-hand reused data”

2. Handling Missing Data

What is Missing Data?

Jab dataset mein kisi field ki value missing ho.

Example

Name	Age
A	20
B	NULL
C	22

Yaha B ki age missing hai.

Causes of Missing Data

- Human error
 - Sensor failure
 - Incomplete forms
 - Data corruption
 - Network problems
-

Problems Due to Missing Data

- Wrong analysis
 - ML accuracy reduce hoti hai
 - Statistical calculations incorrect ho jaate hain
-

Techniques to Handle Missing Data

Technique 1: Remove Rows

Missing value wali rows delete kar do.

Example

Before:

Age
20
NULL
25

After:

Age
20
25

Advantages

- Easy
- Fast

Disadvantages

- Data loss
 - Small dataset aur chhota ho sakta hai
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Technique 2: Mean Imputation

Missing numerical values ko average se replace karte hain.

Formula

Mean = Sum of values / Number of values

Numerical Example

Data: 20, 25, NULL, 35

Mean:

$$(20 + 25 + 35) / 3 = 26.67$$

Replace NULL with 26.67

New Data: 20, 25, 26.67, 35

Advantages

- Maintains dataset size
- Easy to calculate

Disadvantages

- Outliers mean ko affect karte hain
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Technique 3: Median Imputation

Middle value use karte hain.

Useful when outliers present ho.

Numerical Example

Data: 10, 12, 15, 100

Median:

$$(12 + 15) / 2 = 13.5$$

Missing value replaced by 13.5

Advantages

- Outlier resistant

Disadvantages

- Exact data distribution maintain nahi hoti
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Technique 4: Mode Imputation

Most frequent value fill karte hain.

Useful for categorical data.

Numerical Example

Colors: Red, Blue, Red, NULL

Mode = Red

Result: Red, Blue, Red, Red

Technique 5: Forward Fill

Previous row ki value copy karte hain.

Example

Temp
30
NULL
32

After forward fill:

Temp
30
30
32

Pandas Code Example

```
import pandas as pd

df['Age'].fillna(df['Age'].mean(), inplace=True)
```

2. Outliers

What are Outliers?

Aisi values jo baaki data se bahut alag ho.

Example

10, 12, 15, 18, 200

200 = outlier

Causes of Outliers

- Data entry mistakes
 - Sensor error
 - Fraud activity
 - Rare events
-

Problems Due to Outliers

- Average distort ho jata hai
 - Graphs misleading ho jaate hain
 - ML accuracy reduce hoti hai
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Techniques to Detect Outliers

Technique 1: Z-Score Method

Formula

$z = (x - \text{mean}) / \text{standard deviation}$

Where:

- x = value
- mean = average
- standard deviation = spread of data

Rule

If:

- $z > 3$
- $z < -3$

Then outlier.

Numerical Example

Data: 10, 12, 14, 15, 200

Mean ≈ 50 Std deviation ≈ 75

For 200:

$$z = (200 - 50) / 75 = 2$$

Technique 2: IQR Method

Formula

$$\text{IQR} = Q3 - Q1$$

Outlier if:

$$x < Q1 - 1.5(\text{IQR})$$

or

$$x > Q3 + 1.5(\text{IQR})$$

Numerical Example

Data: 10, 12, 14, 15, 100

$$Q1 = 12 \quad Q3 = 15$$

$$\text{IQR} = 15 - 12 = 3$$

Upper limit:

$$15 + 1.5(3) = 19.5$$

$$100 > 19.5$$

Therefore 100 is an outlier.

Handling Outliers

- Remove them
 - Replace with median
 - Cap extreme values
 - Use robust ML algorithms
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Pandas Example

```
df = df[df['Salary'] < 100000]
```

3. Noise

What is Noise?

Random unwanted disturbance in data.

Example

Actual temperature = 30°C

Sensor shows = 90°C

90°C = noisy value

Causes of Noise

- Hardware problems
 - Human mistakes
 - Transmission error
 - Environmental interference
-

Problems Due to Noise

- Wrong predictions
 - Low model accuracy
 - Poor visualization
-

Noise Reduction Techniques

Technique 1: Binning

Data ko groups (bins) mein divide karte hain.

Numerical Example

Data: 2, 4, 5, 20, 22, 24

Bin1: 2, 4, 5

Average:

$$(2 + 4 + 5) / 3 = 3.67$$

Replace all with 3.67

Technique 2: Regression

Best fit line se smoothing karte hain.

Technique 3: Clustering

Noisy points ko cluster se separate kar dete hain.

4. Data Transformation

What is Data Transformation?

Data ko useful format mein convert karna.

Why Needed?

- ML compatibility
 - Better visualization
 - Standard formatting
-

Transformation Techniques

Technique 1: Encoding

Text ko numeric format mein convert karna.

Example

Gender
Male
Female

Converted:

Gender
1
0

Technique 2: Log Transformation

Large values compress karna.

Formula

$\log(x)$

Numerical Example

$x = 1000$

$\log(1000) = 3$

Technique 3: Aggregation

Multiple records combine karna.

Example: Daily sales → Monthly sales

Pandas Example

```
pd.get_dummies(df['Gender'])
```

5. Normalization

What is Normalization?

Different scale wale data ko same range mein lana.

Example Problem

Salary	Age
500000	20

Salary bahut large hai.

ML model salary ko zyada importance dega.

Normalization Techniques

Technique 1: Min-Max Normalization

Formula

$$x' = (x - x_{\min}) / (x_{\max} - x_{\min})$$

Range

0 to 1

Numerical Example

Data: 10, 20, 30

Normalize 20:

$$(20 - 10) / (30 - 10)$$

$$= 10 / 20$$

$$= 0.5$$

Technique 2: Z-Score Normalization

Formula

$$z = (x - \text{mean}) / \text{standard deviation}$$

Numerical Example

Value = 20 Mean = 15 Std deviation = 5

$$z = (20 - 15) / 5 = 1$$

Advantages of Normalization

- Better ML performance
 - Faster training
 - Fair feature comparison
-

6. Data Wrangling

What is Data Wrangling?

Raw data ko clean, organize, and prepare karna.

Also called:

- Data munging
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Tasks in Data Wrangling

- Remove duplicates
 - Handle missing values
 - Filter data
 - Merge datasets
 - Transform columns
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Pandas Library

What is Pandas?

Python library for:

- Data analysis
 - Data manipulation
 - Data cleaning
-

Create DataFrame

```
import pandas as pd

df = pd.DataFrame({
    'Age': [20, None, 25]
})
```

Check Missing Values

```
df.isnull()
```

Remove Missing Values

```
df.dropna()
```

Fill Missing Values

```
df.fillna(df.mean())
```

Filter Data

```
df[df['Age'] > 21]
```

NumPy Library

What is NumPy?

Python library for numerical operations.

Create Array

```
import numpy as np  
  
arr = np.array([1,2,3])
```

Mean

```
np.mean(arr)
```

Median

```
np.median(arr)
```

Standard Deviation

```
np.std(arr)
```

Real-Life Example

Suppose e-commerce dataset:

User	Age	Salary
A	20	50000
B	NULL	60000
C	22	9999999

Problems:

- Missing age
- Salary outlier

Solutions:

- Fill age with mean
- Remove/cap salary outlier
- Normalize salary

Viva Questions

Q1. What is missing data?

Data having empty values.

Q2. What is an outlier?

An abnormal value far from others.

Q3. Why normalization is needed?

To bring data on same scale.

Q4. What is data wrangling?

Cleaning and organizing raw data.

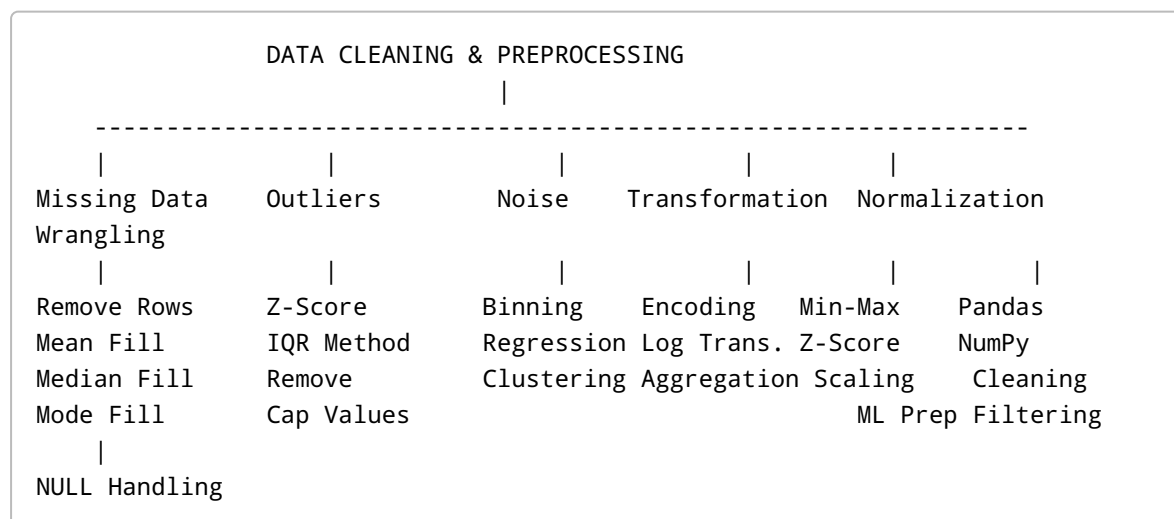
Q5. Which library is commonly used for data cleaning?

Pandas.

Quick Revision Table

Topic	Main Purpose
Missing Data	Handle empty values
Outliers	Detect abnormal values
Noise	Remove random errors
Transformation	Convert format
Normalization	Bring same scale
Wrangling	Overall preprocessing

Complete Mind Map Revision



One-Line Revision

- Missing values → fill/remove

- Outliers → abnormal values
- Noise → random errors
- Transformation → format conversion
- Normalization → same scale
- Wrangling → complete preprocessing